R16

Code No: 131AH

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year I Semester Examinations, May - 2018
ENGINEERING PHYSICS-I
(Common to EEE, ECE, CSE, EIE, IT, ETM)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Explain basic difference between division of amplitude and division of wave front. [2]
b) Describe the properties of Fraunhofer diffraction. [3]
c) Write short notes on double refraction. [2]
d) Discuss about stimulated emission. [3]
e) Explain construction of optical fiber. [2]
f) Discuss about attenuation in optical fibers. [3]
g) Write salient features of miller indices. [2]
h) Calculate packing factor of diamond. [3]
i) Write short notes on Burger's vector. [2]
j) Write short notes on point defects. [3]

PART-B

(50 Marks)

2.a) Explain interference in thin films in transmitted light.
b) Give an account of grating experiment. [5+5]

OR

3.a) Explain Franhofer diffraction due to single slit and extend it to N-slits.
b) Discuss in detail about spatial and temporal coherence. [5+5]

4.a) Discuss about the characteristics of lasers.
b) Describe principle, working and construction of semiconductor lasers. [5+5]

OR

5.a) Give an account of absorption, spontaneous and stimulated emission.
b) Explain principle and working nicol prism.
c) Write any four applications of lasers. [10]

6.a) Explain the use of fiber optic cables in communication system.
b) Obtain an expression for numerical aperture.
c) Write any four applications of fiber optics in medicine. [10]

OR

7.a) Discuss about construction and principle of optical fiber with the help of neat diagrams.
b) Write in detail about step index and graded index fibers. [5+5]

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8.a) Define atomic radius, coordination number and packing fraction.
b) Explain classification of crystals based on the lattice parameters.
c) Write short notes on Miller indices. [10]

OR

9.a) Prove that FCC is closely packed when compared to bcc and sc.
b) Write short notes on HCP and crystal planes and directions. [5+5]

10.a) Discuss about line defects and also explain the significance of Burger's vector.
b) Discuss about determination of crystal structure by Laue method. [5+5]

OR

11.a) Give an account of vacancies, substitutional, interstitial defects.
b) Estimate concentration of Frenkel defects at a given temperature. [5+5]

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